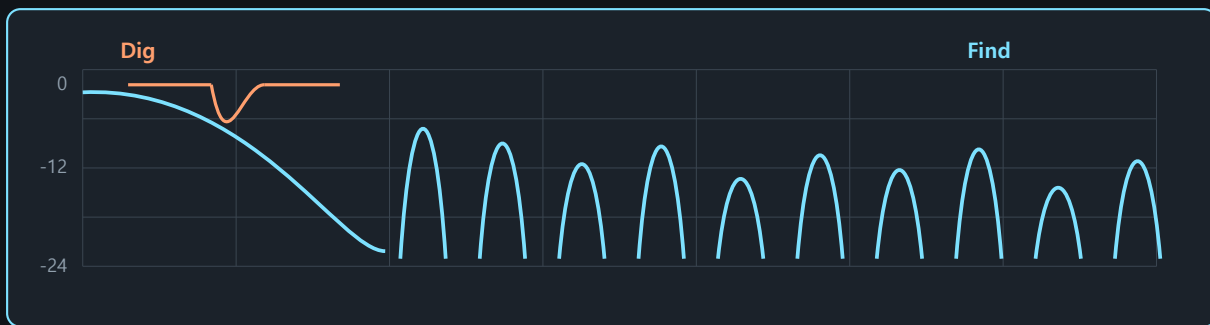


TB_Jewel Digger & Finder

User Manual - Current implementation / 2026-06-30



A musical filter-motion plug-in that combines a scale-aware filter bank with a parallel crystal grain engine. Find emphasizes selected harmonic bands, while Dig carves them out for notch and comb-filter textures.



Core Idea

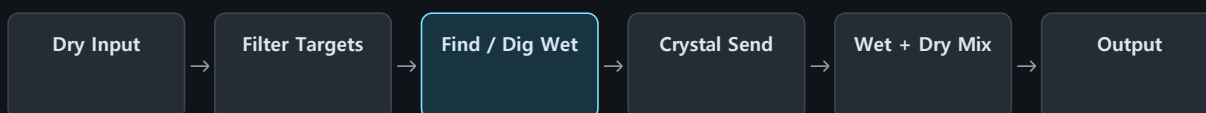
- Base Note selects the musical reference point across the C0-C8 range.
- Scale, Shovel, and Motion create a controlled but animated filter pattern.
- Crystal runs as a parallel send from the filter-bank signal, adding pitch, delay, feedback, and reverse probability.
- Mix blends dry and final wet, while Output trims the final level.

Quick Start

- Set Base Note to the song key or the frequency area you want to organize. The control is snapped and displayed as C0-C8 note names.
- Choose Find or Dig. Find emphasizes harmonic bands, while Dig carves those bands out like moving notches.
- Raise Amount to set the effect intensity. Use Q and Density to shape the width and number of active bands.
- Choose Motion and Rate to animate the filter targets. Enable Sync to make Rate follow DAW-tempo divisions from 1/64 to 1/1.
- Add the crystal grain layer with Crystal Send and Crystal Mix when needed.
- Finish with Mix and Output. Mix 0% is dry, while 100% is wet-focused.

Suggested start: Find, Amount 35%, Q 6-10, Density 12, Motion Arp Up or Sine, Crystal Mix 0-20%, Mix 100%.

Signal Flow



The filter bank selects scale-valid candidate frequencies above the Base Note and spreads Density targets up toward 20 kHz. Shovel changes the level profile and, for some profiles, the fine detune pattern of the bands.

In Find mode with Low Cut off, the low range below the Base Note is protected and passed through. With Low Cut on, the old behavior is restored and the area below the Base Note is cleaned by a 24 dB/oct high-pass. Dig subtracts selected harmonic bands from the dry signal to create notch and comb-filter motion.

Crystal receives a parallel send from the filter-bank signal. Crystal Only removes the filtered wet path and sends only the crystal path to the final wet bus. If Mix is below 100%, dry signal still remains.

Reading the Analyzer

Item	Meaning
Color	Find is cyan, Dig is orange. The graph line and theme follow the selected mode.
X-axis	20 Hz-20 kHz logarithmic frequency view with C0-C8 octave markers.
Y-axis	Displayed from 0 to -24 dBFS with 0, -6, -12, -18, and -24 dB labels.
Find	Shows band-pass-like peaks at selected musical frequencies. With Low Cut off, the protected low range below the Base Note is also shown.
Dig	Shows downward notches at selected targets, reading like a comb/notch response.
Motion	LFO, arpeggio, and random motion update live in the graph.

Filtering Section

Control	Range / Default	Function
Base Note	C0-C8 / default A3	Reference note for the filter bank. Targets are selected above this snapped musical frequency.
Scale	Chromatic, Major, Minor, etc.	Limits target pitch classes. Includes Major 7, Minor 7, Diminished, Augmented, Pentatonic Major/Minor, and Mixolydian.
Shovel	Crystal, Hollow, Bow, Hammer, Metal, Dirt	Changes band weight and selected detune behavior, from smooth/even to irregular and gritty.
Mode	Find / Dig	Find emphasizes selected bands. Dig removes them from the dry signal.
Amount	0-100% / default 35%	Main Find/Dig intensity. Higher values create stronger boosts or deeper notches.
Q	Mode-aware displayed value	Find uses a sharper band range, while Dig uses a notch-width range. The displayed value is the effective mode-scaled Q.
Density	1-24 / default 12	Number of active filter targets. Higher values create a denser pattern up toward 20 kHz.
Rate	0.02-64 Hz or 1/64-1/1	Motion speed. With Sync enabled, the display follows DAW BPM divisions.
Motion Depth	0-24 st / default 7 st	How far Motion moves above or below the reference position.
Random	0-100%	Enabled only for Musical Random or Chaos. Controls how far the motion drifts from ordered scale movement.
Low Cut	Button / default Off	Applies a 24 dB/oct high-pass around the Base Note. In Find with Low Cut off, the lower range is protected.

Motion Modes

Mode	Behavior
Still	Keeps the current Base Note and Scale map without motion.
Arp Up / Arp Down	Steps upward or downward through the selected scale.
Ramp Up / Ramp Down	Linear LFO-style rise or fall, not arpeggiated stepping.
Sine / Square / Triangle	Classic LFO shapes moving above and below the reference point.
Musical Random	Blends scale-aware ordered motion with randomized scale offsets.
Chaos	Less musical random phase/modulation movement for unstable textures.

Crystallizing Section

Control	Range / Default	Function
Crystal Mix	0-100% / default 20%	Final level of the crystal grain layer.
Crystal Send	0-100% / default 80%	How much filter-bank signal feeds the Crystal engine.
Pitch	-24 st to +24 st / default 0 st	Each new grain chooses a random pitch between 0 and the set value. Positive values randomize upward; negative values randomize downward.
Attack	10-750 ms / default 120 ms	Attack time of each grain envelope.
Decay	10-950 ms / default 400 ms	Decay time of each grain envelope.
Delay	0-1000 ms / default 20 ms	Delay read time for the grain buffer.
Feedback	0-100% / default 25%	Regeneration amount in the grain delay path.
Reverse Chance	0-100% / default 0%	Probability that each new grain plays in reverse.
Crystal Only	Button / default Off	Uses only the crystal path as the wet signal. Final dry/wet balance is still controlled by Mix.

Mix / Output Section

Control	Range / Default	Function
Mix	0-100% / default 100%	Blends dry and final wet. 0% is dry only, 100% is wet only.
Output	-18 dB to +6 dB / default 0 dB	Final output level trim.

Practical Tips

- For vocals or leads, use Find with Density around 6-12 and a higher Q for more focused harmonic accents.
- For pads and drones, Dig with Sine or Triangle Motion is useful for slow phasing and comb-filter movement.
- When using Crystal heavily, raise Crystal Send first, then use Crystal Mix to place the layer in the final sound.
- Enable Sync and try Rate around 1/8 to 1/2 for quick tempo-locked motion.
- Think of Low Cut On as cleaning the range below the Base Note, and Low Cut Off as preserving the low foundation in Find mode.